A UNIFORM AVERAGE SENSITIVITY BOUND

Abstract. We give a self-contained reconstruction of the compensated coupling bound of Lagdive, Aglave, and Damre: every real degree-at-most-d polynomial threshold function on the uniform sign cube in n variables has total influence at most $(8/\sqrt{2\pi})d\sqrt{n}$. The proof retains OpenAI's weighted degree filtration, ordered-edge comparison, and parity coupling, and includes the finite-parameter estimate $\min\{n, 8db_n(n+d)/(n+2d)\}$, where $b_n = \mathbb{E}|\operatorname{Bin}(n,1/2) - n/2|$. An elementary Wallis argument handles every dimension, and a rational certificate gives the explicit corollary $\mathrm{AS}(f) \le (399/125)d\sqrt{n}$. Exact arithmetic scripts check the certificate and finite coupling and operator instances. The argument follows the compensated coupling of Lagdive, Aglave, and Damre.

1. Statement, scope, and source comparison

Let $\Omega_n = \{-1,1\}^n$, let X be uniform on Ω_n, and let $x^{\oplus i}$ denote x with its ith coordinate reversed. We use $\operatorname{sgn}(t) = 1$ for $t \ge 0$ and $\operatorname{sgn}(t) = -1$ for $t < 0$. The total influence, or average sensitivity, of $f : \Omega_n \to \{-1,1\}$ is

$$\mathrm{AS}(f) = \sum_{i=1}^{n} \mathbb{P}\{f(X) \ne f(X^{\oplus i})\}. \tag{1}$$

In particular, $\mathrm{AS}(f)$ is the number of *ordered* sensitive edges divided by 2^n. All polynomials below are real; no regularity assumption or asymptotic restriction on d is imposed.

Theorem 1 (Reconstruction of [1], Theorem 1)**.** *For every pair of integers $n, d \ge 1$ and every polynomial p of degree at most d, let $f = \operatorname{sgn} p$ on Ω_n. Set*

$$b_n = \mathbb{E}|B_n - n/2| = \frac{n}{2^n}\binom{n-1}{\lfloor (n-1)/2 \rfloor}, \qquad B_n \sim \operatorname{Bin}(n,1/2). \tag{2}$$

Then

$$\mathrm{AS}(f) \le \min\left\{n,\ 8db_n\frac{n+d}{n+2d}\right\}, \qquad \mathrm{AS}(f) \le \frac{8}{\sqrt{2\pi}}d\sqrt{n}. \tag{3}$$

Consequently the explicitly specified rational $C = 399/125$ satisfies

$$\mathrm{AS}(f) \le Cd\sqrt{n}, \qquad \frac{8}{\sqrt{2\pi}} < \frac{399}{125} < \frac{383}{100} < 8. \tag{4}$$

The coefficient 8 and the operator reduction are due to OpenAI [3]. The earlier note of Lagdive, Aglave, and Damre [2] gave the uniform coefficient 383/100. Their later preprint [1], posted on 13 October 2026, already proves both bounds in (3) for $0 \le d \le n$. The coefficient that was asymptotic in the earlier note is uniform in the later version. The proof below retains the given d even when $d > n$. Constants and zeros of p are included.

The proof has four ingredients: a compensated equal-marginal estimate, the original weighted filtration and edge comparison, a parity-induced binomial coupling, and exact even/odd binomial estimates. None is assumed from an external source.

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2. A compensated equal-marginal estimate

Lemma 2 (Compensated coupling, cf. [1], Lemma 2)**.** *Let U, V have the same distribution and finite second moment, with common mean m. Suppose $V - U \le a$ almost surely, where $a \ge 0$. Write $b = \mathbb{E}|U - m|$, $D = V - U$, and $t = \mathbb{E}D_+ = \mathbb{E}D_-$. Then*

$$\mathbb{E}D^2 + at \le 4ab. \tag{5}$$

If both variables lie in a common interval of length $L > 0$, then

$$\mathbb{E}D^2 \le 4ab\frac{L + a}{L + 2a}. \tag{6}$$

No independence or exchangeability is required.

Proof. Set

$$g(z) = \tfrac{1}{2}(z - m)|z - m| = \int_m^z |s - m|\, ds.$$

For $u < v$, integrating the chord bound for the convex function $|s - m|$ gives the upper bound in

$$\frac{(v - u)^2}{4} \le g(v) - g(u) \le \frac{v - u}{2}(|u - m| + |v - m|). \tag{7}$$

For the lower bound, if $u \le m \le v$, the integral is $((m - u)^2 + (v - m)^2)/2 \ge (v - u)^2/4$. If the interval is on one side of m, the integral is at least $(v - u)^2/2$.

All expectations exist because g has quadratic growth. Equality of the marginals gives

$$\mathbb{E}[(g(U) - g(V))\mathbf{1}_{D<0}] = \mathbb{E}[(g(V) - g(U))\mathbf{1}_{D>0}].$$

Putting $A = |U - m| + |V - m|$, applying the two sides of (7) on the respective events, and using $D \le a$ on $D > 0$, yields

$$\mathbb{E}[D^2\mathbf{1}_{D<0}] \le 4\mathbb{E}[(g(V) - g(U))\mathbf{1}_{D>0}] \le 2a\mathbb{E}[A\mathbf{1}_{D>0}].$$

Since $A \ge |D|$ pointwise and $\mathbb{E}A = 2b$, we retain the complement:

$$\mathbb{E}[A\mathbf{1}_{D>0}] = 2b - \mathbb{E}[A\mathbf{1}_{D<0}] - \mathbb{E}[A\mathbf{1}_{D=0}] \le 2b - \mathbb{E}D_- = 2b - t.$$

Also $\mathbb{E}[D^2\mathbf{1}_{D>0}] \le a\mathbb{E}D_+ = at$. Adding proves $\mathbb{E}D^2 \le 2a(2b - t) + at = 4ab - at$. This includes $a = 0$.

For the finite-range assertion, $D_- \le L$ and $D_+ \le a$, so $Q := \mathbb{E}D^2 \le (L + a)t$. Multiply $Q + at \le 4ab$ by $L + a$ and add a times $Q - (L + a)t \le 0$. The result is $(L + 2a)Q \le 4ab(L + a)$; division is valid since $L > 0$. □

3. The weighted degree filtration

On the cube, replacing each x_i^2 by 1 makes p multilinear without raising its degree or changing its values. It then has degree at most $\min(d, n)$. Add a positive constant smaller than the absolute value of every negative value of p. If there are no negative values, any positive constant suffices. The finite cube makes this choice possible. It preserves $\operatorname{sgn} p$ with the specified tie convention, does not increase the degree, and makes p nonzero everywhere. We henceforth use this polynomial.

Put $N = 2^n$ and use the counting inner product on $\mathcal{H} = \mathbb{C}^{\Omega_n}$, so the point masses form an orthonormal basis. For an operator B let $\|B\|_{\mathrm{HS}}^2 = \sum_{x,y} |B_{xy}|^2$. The characters $\chi_S(x) = \prod_{i \in S} x_i$ are an orthogonal basis: a nonconstant character sums to zero by pairing points across one of its coordinates. Also $\chi_S\chi_T = \chi_{S \triangle T}$.

Let V_k be the span of characters with $|S| \le k$, taking $V_k = \mathcal{H}$ for $k \ge n$. Fix a positive weight w, and let D_w multiply by $\sqrt{w}$. Set

$$K_k = D_w V_k, \qquad K_{-1} = \{0\}, \qquad E_k = K_k \cap K_{k-1}^{\perp}.$$

The spaces are nested and D_w is invertible. Thus $\mathcal{H} = \bigoplus_{k=0}^{n} E_k$ orthogonally and $\dim E_k = \binom{n}{k}$. Write Π_k for projection onto E_k, and put

$$M = \sum_{k=0}^{n} (k - n/2)\Pi_k.$$

This self-adjoint operator satisfies

$$\|M\|_{\mathrm{HS}}^2 = \sum_k \binom{n}{k}(k - n/2)^2 = \frac{nN}{4}. \tag{8}$$

The sum is N times the variance of a sum of n independent fair Bernoulli variables, whose variance is $n/4$.

Let Z_i multiply by coordinate x_i. It is a self-adjoint unitary, commutes with D_w, and sends K_r into K_{r+1}. Hence $\Pi_s Z_i \Pi_r = 0$ if $s > r + 1$; taking adjoints gives

$$\Pi_s Z_i \Pi_r = 0 \quad \text{if } |s - r| > 1. \tag{9}$$

Orthogonality of both the domain and range decompositions also gives, for every operator B,

$$\|B\|_{\mathrm{HS}}^2 = \sum_{r,s} \|\Pi_s B \Pi_r\|_{\mathrm{HS}}^2. \tag{10}$$

4. Recovering ordered edges

Let $h : \Omega_n \to \{-1, 1\}$, let H_h multiply by h, and let $\mathcal{E}(h)$ be the set of ordered neighboring pairs (x, y) for which $h(x) = h(y)$.

Lemma 3 (Edge comparison from [3])**.** *For every positive weight w and every such h,*

$$|\mathcal{E}(h)| \le 2\|MH_h + H_h M\|_{\mathrm{HS}}^2. \tag{11}$$

Proof. Let $C_i = [M, Z_i]$, whose (s, r) block is $(s - r)\Pi_s Z_i \Pi_r$. The unitaries

$$J = \sum_k (-1)^k \Pi_k, \qquad W = \sum_k \mathrm{i}^k \Pi_k$$

show that $O_i = (Z_i - JZ_iJ)/2$ consists of the adjacent-grade blocks of Z_i, by (9). Its operator norm is at most one. On those blocks $\mathrm{i}^{s-r} = \mathrm{i}(s - r)$, so $WO_iW^* = \mathrm{i}C_i$ and $\|C_i\|_{\mathrm{op}} \le 1$. In the point basis, $(C_i)_{xy} = (y_i - x_i)M_{xy}$. Choosing a differing coordinate for $x \ne y$ gives $|M_{xy}| \le 1/2$, because each matrix entry is bounded by the operator norm. Every edge deficit $1/4 - |M_{xy}|^2$ is therefore nonnegative.

By (9) and (10),

$$\|C_i\|_{\mathrm{HS}}^2 = \sum_{r,s} (s - r)^2 \|\Pi_s Z_i \Pi_r\|_{\mathrm{HS}}^2 \le \|Z_i\|_{\mathrm{HS}}^2 = N.$$

Writing ρ for Hamming distance and summing point-basis entries yields

$$\sum_{x,y} \rho(x, y)|M_{xy}|^2 = \tfrac{1}{4}\sum_i \|C_i\|_{\mathrm{HS}}^2 \le \frac{nN}{4} = \sum_{x,y} |M_{xy}|^2. \tag{12}$$

Set $A_0 = \sum_x |M_{xx}|^2$ and $F = \sum_{\rho(x,y)\ge 2} |M_{xy}|^2$. Subtracting the extreme sums in (12) gives $F \le \sum_{\rho \ge 2}(\rho - 1)|M_{xy}|^2 \le A_0$. Since there are nN ordered cube edges, (8) gives

$$\sum_{\rho(x,y)=1} (1/4 - |M_{xy}|^2) = A_0 + F \le 2A_0. \tag{13}$$

Let $A_h = \sum_{(x,y)\in\mathcal{E}(h)} |M_{xy}|^2$. Nonnegativity of the deficits implies $|\mathcal{E}(h)|/4 \le A_h + 2A_0 \le 2(A_h + A_0)$. The entries of $T = MH_h + H_hM$ are $(h(x) + h(y))M_{xy}$. Their squared factor is four on the diagonal and on $\mathcal{E}(h)$, so $\|T\|_{\mathrm{HS}}^2 \ge 4(A_h + A_0)$. Combining proves (11). □

5. Parity and the binomial coupling

Choose $w = |p| > 0$, put $\chi = \chi_{[n]}$, and let $h = \chi f$ and $q = \chi p$. Then $hw = q$. Full parity flips across every cube edge, so $\mathcal{E}(h)$ is precisely the set of ordered sensitive edges of f:

$$\mathrm{AS}(f) = |\mathcal{E}(h)|/N. \tag{14}$$

Multiplication by χ complements the support of each character, so q has Fourier support only in degrees at least $n - d$. If $u \in V_r$ and $v \in V_s$, then $\overline{v}u$ has degree at most $r + s$, giving

$$\langle D_w v, H_h D_w u\rangle = \sum_x \overline{v(x)}u(x)q(x) = 0 \quad \text{if } r + s < n - d.$$

Since $E_r \subseteq D_w V_r$, this implies

$$\Pi_s H_h \Pi_r = 0 \quad \text{if } r + s < n - d. \tag{15}$$

Define a joint law by

$$\mathbb{P}(R = r, S = s) = N^{-1}\|\Pi_s H_h \Pi_r\|_{\mathrm{HS}}^2.$$

Unitarity of H_h and (10) give

$$\sum_s \|\Pi_s H_h \Pi_r\|_{\mathrm{HS}}^2 = \|H_h \Pi_r\|_{\mathrm{HS}}^2 = \binom{n}{r},$$

and summing over r similarly gives $\binom{n}{s}$. The masses sum to one, and both marginals are $\mathrm{Bin}(n, 1/2)$. For $T = MH_h + H_h M$,

$$\Pi_s T \Pi_r = (r + s - n)\Pi_s H_h \Pi_r, \qquad N^{-1}\|T\|_{\mathrm{HS}}^2 = \mathbb{E}(R + S - n)^2.$$

Set $U = S$ and $V = n - R$. Binomial symmetry gives identical marginals for U, V, and (15) gives $V - U \le d$. If $d > n$, this last constraint is automatic and the construction still applies. The edge comparison and (14) therefore give

$$\mathrm{AS}(f) \le 2\mathbb{E}(U - V)^2 \le 8db_n \frac{n + d}{n + 2d}, \tag{16}$$

where the last inequality is Lemma 2 with $a = d$ and common support $[0, n]$. Also each summand in (1) is at most one, so $\mathrm{AS}(f) \le n$. This proves the finite bound.

6. An all-dimension binomial estimate

Symmetry and the identity

$$(2j - n)\binom{n}{j} = n\left[\binom{n-1}{j-1} - \binom{n-1}{j}\right]$$

give, on summing from $j_0 = \lfloor n/2\rfloor + 1$ to n,

$$b_n = 2^{-n}\sum_{j=j_0}^{n}(2j - n)\binom{n}{j} = n2^{-n}\binom{n-1}{j_0 - 1}.$$

For even n, the two central coefficients in row $n - 1$ are equal, proving the stated formula (2). With $c_k = 4^{-k}\binom{2k}{k}$,

$$b_{2k} = kc_k \quad (k \ge 1), \qquad b_{2k+1} = \frac{2k + 1}{2}c_k \quad (k \ge 0). \tag{17}$$

We include an elementary Wallis estimate to avoid any unproved asymptotic substitution. Let $I_r = \int_0^{\pi/2} \sin^r t\, dt$. Integration by parts gives $I_{r+2} = (r + 1)I_r/(r + 2)$ for $r \ge 0$, with $I_0 = \pi/2$ and $I_1 = 1$. Thus $I_{2k} = \pi c_k/2$ and $I_{2k+1} = 1/((2k + 1)c_k)$. The inequalities $I_{2k+2} \le I_{2k+1} \le I_{2k}$ imply

$$\frac{2}{\pi(2k + 1)} \le c_k^2 \le \frac{2(2k + 2)}{\pi(2k + 1)^2}.$$

Consequently $(k+1/4)c_k^2 \to 1/\pi$. Since $c_{k+1}/c_k = (2k+1)/(2k+2)$ and

$$(2k+1)^2(k+5/4) - (2k+2)^2(k+1/4) = 1/4,$$

the sequence $(k+1/4)c_k^2$ increases. It cannot exceed its limit, so

$$c_k^2 \le \frac{1}{\pi(k+1/4)} \quad (k \ge 0). \tag{18}$$

Combining (17) and (18) gives

$$\frac{2\pi b_n^2}{n} \le \begin{cases} 2n/(2n+1), & n \text{ even}, \\ 2n/(2n-1), & n \text{ odd}. \end{cases} \tag{19}$$

These estimates also imply $b_n^2 \le (n+1)/(2\pi)$: in the odd case the needed inequality is $2n^2 \le (n+1)(2n-1)$, equivalent to $n \ge 1$.

For even n, (19) and $(n+d)/(n+2d) \le 1$ prove the uniform coefficient in (3). For odd n, $d \ge 1$ gives

$$\frac{n+d}{n+2d} \le \frac{n+1}{n+2},$$

because the difference after cross multiplication is $n(d-1) \ge 0$. Finally,

$$(2n-1)(n+2)^2 - 2n(n+1)^2 = 3n^2 + 2n - 4 = (n-1)(3n+5) + 1 > 0.$$

Thus $\sqrt{2\pi}b_n(n+d)/(\sqrt{n}(n+2d)) \le 1$ for odd n as well, including $n = 1$. Equation (16) proves the uniform bound for every $n, d \ge 1$.

7. The rational witness and strict comparisons

For $0 < x \le 1$, integration of the exact identity

$$\frac{1}{1+t^2} = 1 - t^2 + t^4 - t^6 + \frac{t^8}{1+t^2}$$

shows that $\arctan x > x - x^3/3 + x^5/5 - x^7/7$. The sum $\arctan(1/2) + \arctan(1/3)$ lies in $(0, \pi/2)$ and has tangent one, so it equals $\pi/4$. Therefore

$$\pi > 4 \sum_{x \in \{1/2, 1/3\}} \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} \right) = \frac{1538665}{489888} > \frac{500000}{159201} = \frac{32}{(399/125)^2}. \tag{20}$$

The second strict gap is exactly $206455/1237946976$. Squaring positive quantities in (20) yields $8/\sqrt{2\pi} < 399/125$. Also

$$\frac{383}{100} - \frac{399}{125} = \frac{319}{500} > 0, \qquad 8 - \frac{399}{125} = \frac{601}{125} > 0.$$

This proves the rational contract corollary. The stronger uniform bound is retained to make the exact match to [1] explicit.

8. Exact checks and source contributions

The exact checkers in the mathematical supplement verify the rational certificate, binomial identities and finite coupling instances. The parameter-wide inequalities follow from the proofs above.

The weighted-grading construction and its edge and parity reductions are from OpenAI [3]. The compensated inequality and uniform bounds are due to Lagdive, Aglave, and Damre [1]. Their earlier work [2] supplies the intermediate coefficient $383/100$. The rational corollary $399/125 = 3.192$ improves OpenAI's original coefficient 8 for every dimension and degree.

References

[1] Prathmesh Lagdive, Vaibhav Aglave, and Suraj Damre. A compensated coupling bound for the average sensitivity of polynomial threshold functions. https://doi.org/10.21203/rs.3.rs-11323154/v1, October 2026. Research Square preprint, version 1, posted 13 October 2026; manuscript dated 10 October 2026.

[2] Prathmesh Lagdive, Vaibhav Aglave, and Suraj Damre. An elementary improvement of the average sensitivity bound for polynomial threshold functions. https://doi.org/10.5281/zenodo.23269472, October 2026. Zenodo issued 9 October 2026; manuscript dated 10 October 2026.

[3] OpenAI. Average sensitivity of polynomial threshold functions. Official preprint, pinned source, September 2026. 25 September; pinned commit 1581d4cbdd69a0faf34b2089614fe2a814316a6a.

Mathematical supplement

The exact arithmetic programs are included in the attached mathematical supplement (ZIP). Open the attachment in a PDF reader with file-attachment support.